WEEK 2 · DAY 2 · OVERVIEW

Day Counts and Accrued Interest



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ABOUT 45 MIN WITH EXERCISES / 8 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

We buy 1,000,000 of Quorvane Energy, settling Monday 30 November, two months into its coupon period. The seller wants paying for the interest it earned: how much per 100 and in dollars, and why does a calendar give another number? Put the day counts and accrued interest in bondmath, tested against Excel. An assistant has offered to tidy that code; read its change before you take it. Last, show me the holdings file's accrued column is right for all 200 bonds.

Your Route Through Day 2

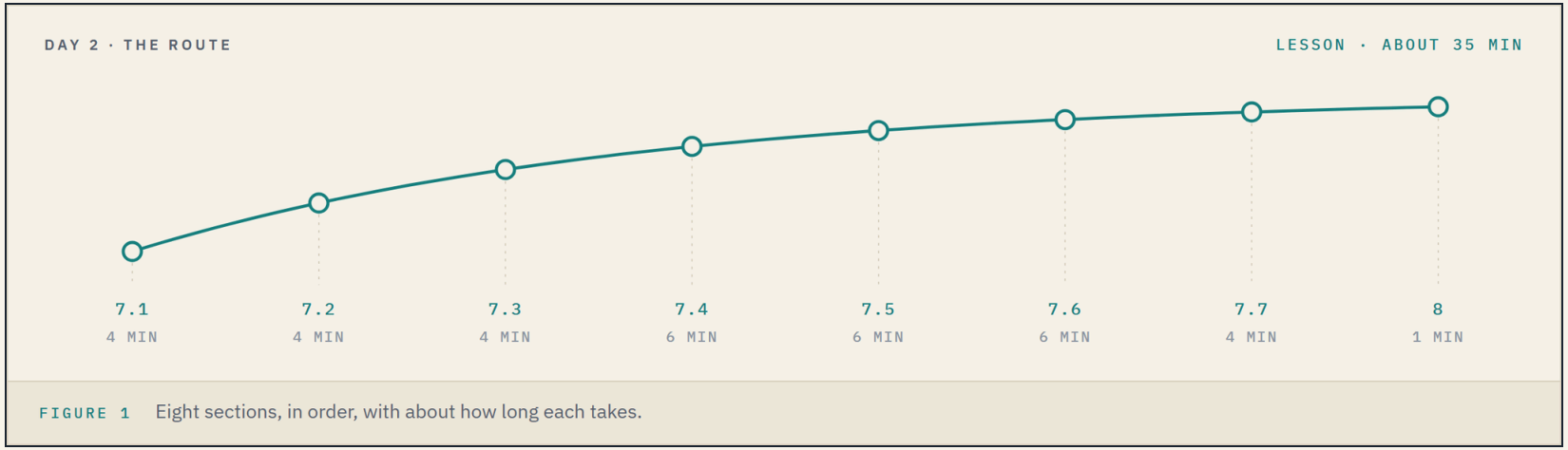


FIGURE 1 Eight sections, in order, with about how long each takes.

	SECTION	ABOUT
7.1	Why the Buyer Pays Accrued Interest	4 min
7.2	The 30/360 Rules, One at a Time	4 min
7.3	**`days_30_360`**	4 min
7.4	A, E, and DSC: **`coupon_period_days`**	6 min

	SECTION	ABOUT
7.5	**`accrued_interest`**	6 min
7.6	Read a Suggested Change Word by Word	6 min
7.7	The Holdings File's **`accrued`** Column	4 min
8	Bring It Home	1 min

Time. About 35 minutes for the lesson, then about 12 for the four exercises.

7.1 Why the Buyer Pays Accrued Interest

COUNT CALENDAR DAYS

```
previous_coupon = date(2026, 9, 30)
next_coupon = date(2027, 3, 31)
days_held = (settlement - previous_coupon).days
days_in_period = (next_coupon - previous_coupon).days
accrued_actual = 5.25 / 2 * days_held / days_in_period
print(f'{days_held} of {days_in_period} days: accrued {accrued_actual:.6f} per 100')
```

```
61 of 182 days: accrued 0.879808 per 100
```

THE FOUR BONDS

	TICKER	MATURITY
0	QVNE	2031-09-30
1	DNMR	2033-09-30
2	PLWK	2029-09-30
3	HLVF	2036-09-30

accrued = coupon_pct / frequency x A / E

7.2 The 30/360 Rules, One at a Time

US 30/360: four rules move D1 or D2, then the formula counts

```
Days = 360 x (Y2 - Y1) + 30 x (M2 - M1) + (D2 - D1)
```

Rule	Case	Day numbers	Days
1 start is the last day of February: D1 = 30	28 Feb 2027 to 1 Mar 2027	D1 28 to 30	**3 to 1**
2 start and end both last of February: D2 = 30	28 Feb 2027 to 29 Feb 2028	D2 29 to 30	**359 to 360**
3 D1 is 31: D1 = 30	31 Aug 2026 to 30 Sep 2026	D1 31 to 30	**29 to 30**
4 D2 is 31 and the start date's day is 30 or 31: D2 = 30	30 Jul 2026 to 31 Oct 2026	D2 31 to 30	**91 to 90**

Rule 4 needs a start on the 30th or 31st: 15 Oct to 31 Oct 2026 keeps D2 = 31, so 16 days.

FIGURE 3 Four rules change a 30/360 count: 28 Feb to 1 Mar 2027 goes from 3 days to 1, and 30 Jul to 31 Oct 2026 from 91 to 90.

THE FORMULA WITH NO RULES

```
2026-08-16 to 2026-09-30: 44 days
2026-08-31 to 2026-09-30: 29 days
```

EXCEL SIDE BY SIDE

JOB	RESULT
The formula, start in A1 and end in B1	44, then 29
Rule 3 in the sheet	30

7.3 days_30_360

THE FOUR RULES

```
    if start_is_feb_end:
        d1 = 30
    if start_is_feb_end and end_is_feb_end:
        d2 = 30
    if start.day == 31:
        d1 = 30
    if end.day == 31 and start.day >= 30:
        d2 = 30
```

THREE FEBRUARY MONTH ENDS

```
2027-02-28 to 2028-02-29 360
2026-02-28 to 2026-10-31 241
2027-02-28 to 2027-02-28 0
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Rule 2	`=YEARFRAC(DATE(2027,2,28),DATE(2028,2,29),0)*360`, then `=DAYS360(DATE(2027,2,28),DATE(2028,2,29))`	360; DAYS360 359
Rule 4 after a February start	`=YEARFRAC(DATE(2026,2,28),DATE(2026,10,31),0)*360`, then `DAYS360`	241; DAYS360 240
The same date twice	`=DAYS360(DATE(2027,2,28),DATE(2027,2,28))`	0; DAYS360 -2

`days_30_360` **counts as** `COUPDAYBS` **and** `YEARFRAC(..., 0)` **do;** `DAYS360` **does not, on February month ends.**

7.4 A, E, and DSC: coupon_period_days

QUORVANE ON BOTH BASES

```
30/360 (60, 180, 120)
ACT/ACT (61, 182, 121)
Excel, basis 0: 60 180 120
```

WHERE COUPDAYSNC IS NOT E - A

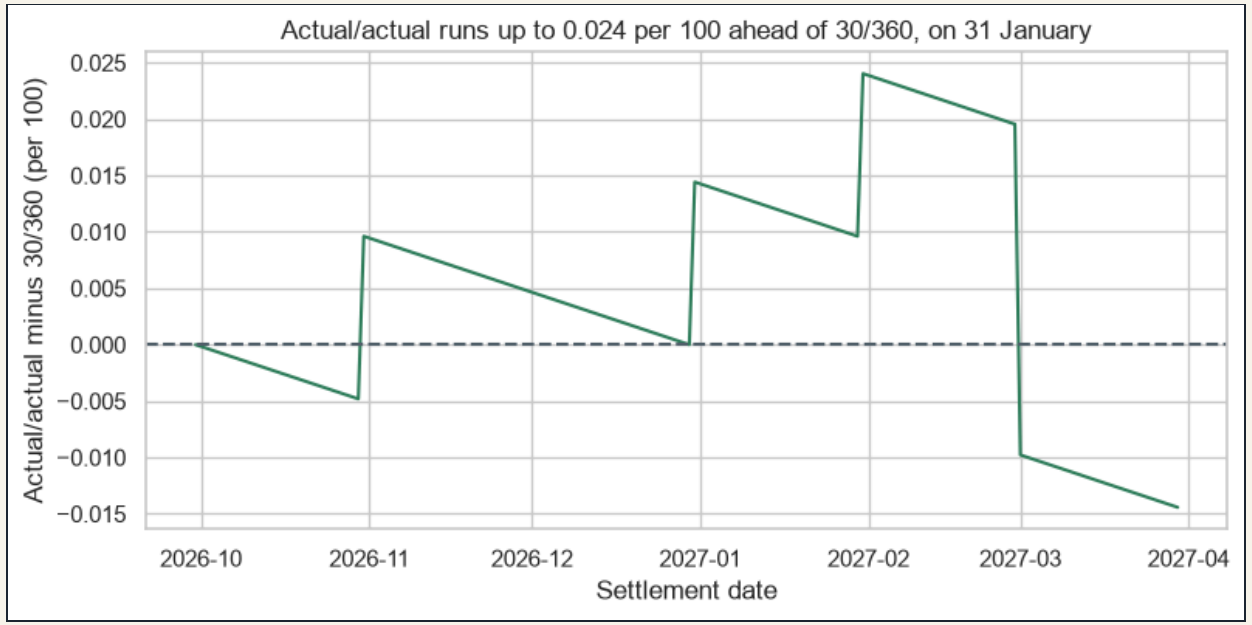
```
278 basis 0 rows; 7 where COUPDAYSNC is not E - A
```

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
A	`=COUPDAYBS(DATE(2026,11,30),DATE(2031,9,30),2,0)`	60
E	`=COUPDAYS(DATE(2026,11,30),DATE(2031,9,30),2,0)`	180
DSC	`=COUPDAYSNC(DATE(2026,11,30),DATE(2031,9,30),2,0)`; `COUPDAYBS(...)+COUPDAYSNC(...)`	120; 180

A, E, and DSC are `COUPDAYBS`, `COUPDAYS`, and the E - A that `PRICE` uses, on every reference row.

7.5 accrued_interest



FOUR BONDS AGAINST EXCEL

	TICKER	PYTHON	EXCEL
0	QVNE	0.875000	0.875000
1	DNMR	1.020833	1.020833
2	PLWK	0.791667	0.791667
3	HLVF	1.250000	1.250000

ACCRINT, THE SECOND ROUTE

```
ACCRINT blank (#NUM! in Excel): 21  different: 9
```

`accrued_interest` **is Excel's accrued formula on all 306 rows, both bases: the desk pays the seller 8,750.00 dollars.**

7.6 Read a Suggested Change Word by Word

Reading git diff --word-diff: only the words that changed are marked

```
{+# a year has 365 days+}
days_in_period = [-360-]{+365+} // frequency
days_to_next = days_in_period - days_accrued
```

`{+ +}` added

`[- -]` removed

no brackets: untouched

One number changed inside one line: 360 became 365, and a comment came with it.

FIGURE 5 Git's word diff marks 360 as removed and 365 as added in the line days_in_period = 360 // frequency.

WORD BY WORD

```
!git -C "{work}" diff --word-diff bondmath.py
```

THROW IT AWAY

```
!git -C "{work}" restore bondmath.py

Accrued again: 0.875000
```

Read every suggested change word by word and let the tests from Excel judge it: 360 is the convention, not a typo.

7.7 The Holdings File's accrued Column

RECOMPUTE ALL 200

```
holdings = pd.read_csv('holdings.csv')
holdings['accrued_python'] = [
    bondmath.accrued_interest(date.fromisoformat(a), date.fromisoformat(m), c)
    for a, m, c in zip(holdings['as_of_date'], holdings['maturity'], holdings['coupon'])]
holdings['gap'] = (holdings['accrued_python'] - holdings['accrued']).abs()
print(f"Largest gap: {holdings['gap'].max():.10f} per 100")
```

CHECK THE LIMIT

```
assert holdings['gap'].max() <= 0.0005 + 1e-9, 'a bond is off by more than rounding'
print('Bonds at a rounding tie:', ((holdings['gap'] - 0.0005).abs() < 1e-9).sum())
```

THE FILE'S ROUNDING

Rounding to 3 decimals moves a number by at most 0.0005, half of the last decimal.

Bring It Home

COPY, TEST, COMMIT

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_07_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py .gitignore
git commit -m "Notebook 07: days_30_360, coupon_period_days, and accrued_interest"
```

Keep the habit: read any suggested edit with `git diff --word-diff`, run `python -m pytest -q`, and `git restore` it if a test fails.

You Can Now

- Compute accrued interest as one coupon times A / E, and say why the buyer pays it.
- Count days on US 30/360 with **`days_30_360`** or **`YEARFRAC(..., 0)`**, and say where **`DAYS360`** differs.
- Count a period on actual/actual, with E as **`COUPNCD - COUPPCD`**.
- Return A, E, and DSC with **`coupon_period_days`**, tied to Excel's **`COUP`** functions.
- Compute accrued with **`accrued_interest`**, and say when **`ACCRINT`** differs.
- Read a suggested change with **`git diff --word-diff`**, judge it with pytest, and discard it with **`git restore`**.
- Explain a gap of exactly half a file's last decimal.

NEXT
WEEK 2 · DAY 3

Clean and Dirty Price

Price a bond between coupons: the quote, the invoice, and a price and a yield for any settlement date, tied to Excel's PRICE and YIELD.